1 2 3 4 5 6 7 8

A B C D

ET0000

Boost_Converter_V1.0

	PCB REF. : Boost_Converter_V1.0	OSM REF. : Boost_Converter_V1.0_01		INDEX : B
	Name	Position	Date	Signature
WRITTEN BY	MHZ	Electronics Designer		
CHECKED BY	MHZ	Electronics Designer		
CHECKED BY				
APPROVED BY	MHZ	Electronics Dept Manager		

CONFIDENTIAL

DIFFUSION :	MHZ

1 2 3 4 5 6 7 8

A B C D

1 2 3 4 5 6 7 8

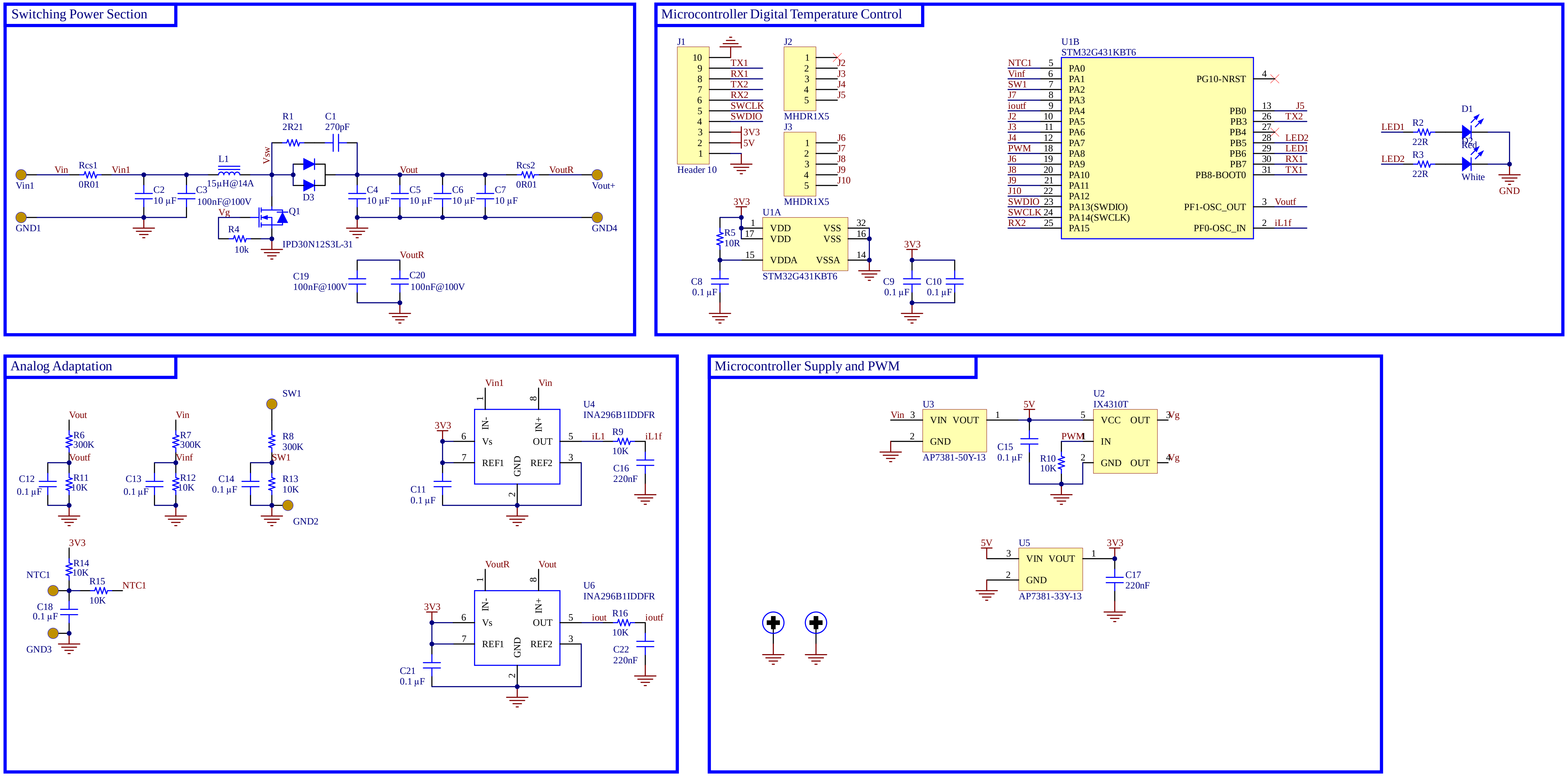
History

ASM REF. : Boost_Converter_V1.0_01_B			
DATE	INDEX	MODIFICATIONS	BY
30/10/2025	A	Second Release, Input and output to 100V	MH

A B C D

A B C D

1 2 3 4 5 6 7 8



Project	ET0000	Date	07/11/2025
Board	Boost_Converter_V1.0		
ASM REF. : Boost_Converter_V1.0_01	ASM INDEX : B	Sheet 3 Of 3	
Drawn by MHZ	Verified by MHZ	Approved by MHZ	

BILL OF MATERIAL

ET0000

Boost_Converter_V1.0

PCB Reference: Boost_Converter_V1.0

Assembly Reference: ASM number : Boost_Converter_V1.0

ASM Variant : 01

ASM Index (Lot) : B

Report Date: 07/11/2025 07:42

#	Designator	Comment	Description	Footprint	manufacturer	Part Number	Supplier 1	Supplier Part Number 1	Code Achat	Quantity
1	C1	270pF	CAP CER 270PF 100V C0G/NP0 0603	CAP 0603_1608	KEMET	C0603C271J1GACAUTO				1
2	C2, C4, C5, C6, C7	10 µF	CAP CER 10UF 100V X7S 1210	MURATA - GRM32E	Murata Electronics	GRM32EC72A106KE05L	DigiKey	490-16266-1-ND		5
3	C3, C19, C20	100nF@100V	CAP CER 0.1UF 100V X8R 0603	CAP 0603_1608	Murata Electronics	GCG188R92A104KA03D	DigiKey	490-18197-1-ND		3
4	C8, C9, C10, C11, C12, C13, C14, C15, C18, C21	0.1 µF	CAP CER 0.1UF 16V X7R 0603	CAP 0603_1608	Yageo	CS0603KRX7R7BB104	DigiKey	311-3560-1-ND		10
5	C16, C17, C22	220nF	CAP CER 0603 220NF 16V X7R 10%	CAP 0603_1608	KEMET	C0603C224K4RECAUTO				3
6	D1	Red	LED RED SMD	LED 0603_1608 RED	Citizen	CL-195HR-CD-T	DigiKey	1642-1492-1-ND		1
7	D2	White	LED WHITE SMD	LED 0603_1608	Citizen	CL-198N-WS-D-T	DigiKey	1642-1495-1-ND		1
8	D3	V40PW10C-M3/I	Rectifier Diode Schottky 100V 40A 3-Pin(2+Tab) Slim DPAK T/R	TO-252	Vishay	V40PW10C	Mouser	78-V40PW10C-M3/I		1
9	L1	15µH@14A	FIXED IND 15UH 14A 9 MOHM SMD	WURTH WE-HCI 1890	Würth Elektronik	74435571500				1
10	Q1	IPD30N12S3L-31	Mosfet Single, 30A, 120V, 57W, To-252-3 Rohs Compliant: Yes |Infineon	TO-252	Infineon	IPD30N06S4L-23	Mouser	726-IPD30N12S3L31AT		1
11	R1	2R21	RES 2.21 OHM 1% 1/8W 0603	RES 0603_1608	KOA Speer Electronics, Inc.	RK73H1JTTD2R21F				1
12	R2, R3	22R	RES SMD 22 OHM 1% 1/10W 0603	RES 0603_1608	Panasonic Electronic Components	ERJ-3EKF22R0V	DigiKey	P22.0HCT-ND		2
13	R4, R9, R10, R11, R12, R13, R14, R15, R16	10k	RES 10K OHM 0.1% 1/10W 0603	RES 0603_1608	Vishay Dale	TNPW060310K0BETA				9
14	R5	10R	RES SMD 10 OHM 1% 1/4W 0603	RES 0603_1608	Rohm Semiconductor	ESR03EZPF10R0	DigiKey	RHM10.0ADCT-ND		1
15	R6, R7, R8	300K	RES SMD 300K OHM 1% 1/3W 0603	RES 0603_1608	Vishay Dale	CRCW0603300KFKEAHP	DigiKey	541-CRCW0603300KFKEAHPCT-ND		3
16	Rcs1, Rcs2	0R01	RES 0.01 OHM 1% 3W 2512	RES 2512_6432	Vishay Dale	WSLP2512R0100FEA				2
17	U1	STM32G431KBT6	STMICROELECTRONICS STM32G431KBT6 MCU ARM, STM32 Family STM32G4 Series Microcontrollers, ARM Cortex-M4F, 32 bits, 170 MHz, 128 KB Fabricant	STM-LQFP32_N	STM	STM32G431KBT6	Farnell	3132405		1
18	U2	IX4310T	2 Ampere High Speed Low-Side MOSFET Driver	SOT-23-5	IXYS	IX4310T				1
19	U3	AP7381-50Y-13	IC REG LINEAR 5V 150MA SOT89	DIODES INC SOT-89	Diodes Incorporated	AP7381-50Y-13				1
20	U4, U6	INA296B1IDDFR	10V/V "-5-V TO 110-V, BIDIRECTIONAL, 1." Current Sense Amplifier 1 Circuit Single-Ended 8-VSSOP	SOT65P280X110-8N	Texas Instruments	INA296B1IDDFR	Mouser	595-INA296B1IDDFR		2
21	U5	AP7381-33Y-13	IC REG LINEAR 3.3V 150MA SOT89	DIODES INC SOT-89	Diodes Incorporated	AP7381-33Y-13				1
										51

Approved

Notes

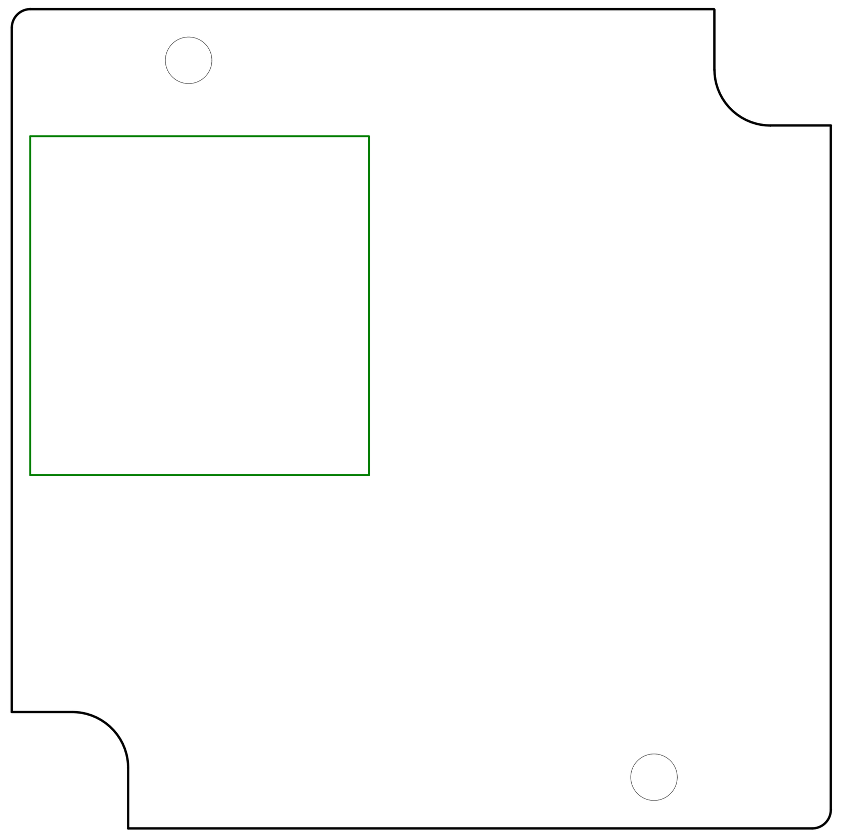
NC/DNC = Non Câblé

Affichage : Verifier l'alignement automatique des lignes sous peine d'occultation de certaines références

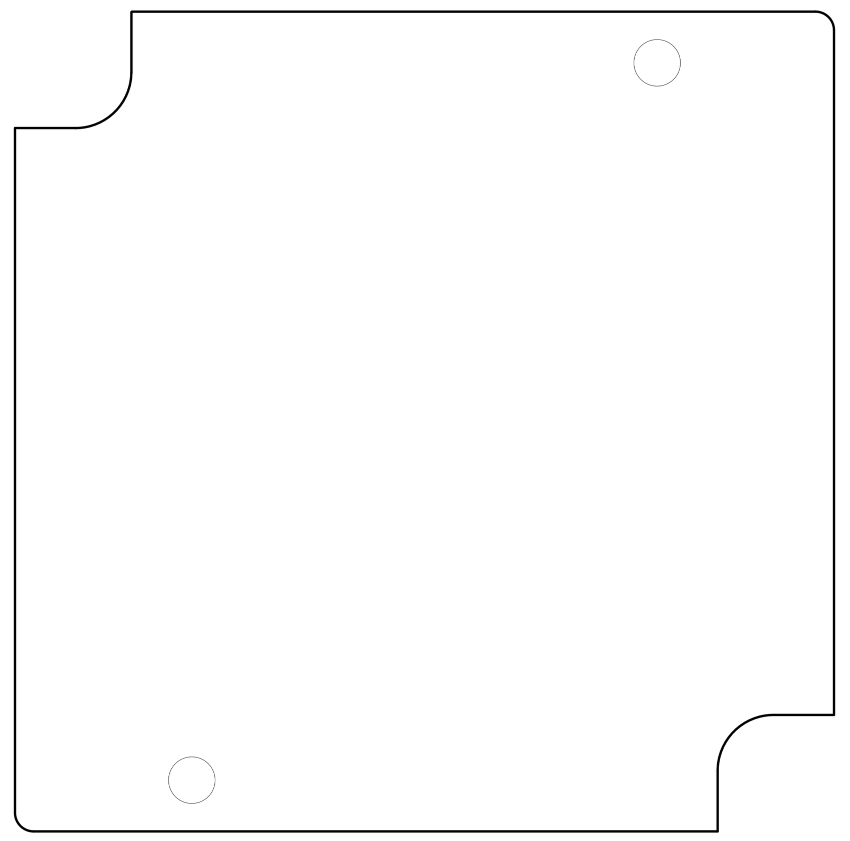
Codes Achat :

A = Appro sur caratéristiques

B = Référence Fabricant à respecter



PROJECT: **ET0000**	PCB REF: Boost_Converter_V1.0	
BOARD: **Boost_Converter_V1.0**	DATE: 07/11/2025	LAYER: Top Assembly



PROJECT:	ET0000	PCB REF:	Boost_Converter_V1.0
BOARD:	Boost_Converter_V1.0	DATE: 07/11/2025	LAYER: Bottom Assembly